

Paper C-03: Trade Networks: A Constraint-Driven Flux Dynamics (CDFD) Perspective

Steve Bico Mujjabi, MD

Independent Researcher

Founder, Vura Labs

Kampala, Uganda

ORCID: 0009-0001-0556-5516

May 2026

Abstract

This paper develops a falsifiable CDFD/AFL model of Trade Networks. It maps goods, services, capital, information, and contractual demand to driving flux Φ , ports, tariffs, logistics, finance, and network dependencies to active constraint C , substitution, inventory, rerouting, diplomacy, and production change to responsiveness S , and contracts, specialization, infrastructure, and disruption history to structural memory M_s . The target outcome is trade volume, shortages, prices, and recovery, tested against customs data, input-output tables, shipping traces, prices, and disruption records. The paper treats universality as a shared model architecture rather than an assertion that unlike systems have the same material mechanism. It states the negative result that would make the mapping unnecessary and places the domain claim inside the cumulative CDFD programme and its reproducible runtime boundary.

Symbol Conventions

CDFD/AFL Part IV symbol convention.

Symbol	Part IV meaning	Cross-domain reading
Φ	Driving flux or throughput	Energy, matter, signal, capital, information, traffic, force, or biological load entering a system
C	Active constraint or capacity burden	Bottleneck, resistance, boundary, scarcity, toxicity, regulation, topology, latency, or load-bearing limit
S	Responsiveness or adaptive surface	The system's ability to reconfigure, buffer, route, repair, learn, or preserve function under changing load
M_s	Structural memory	Stored damage, hysteresis, inherited topology, institutional memory, trained weights, ecological legacy, or retained state
Ψ_s	Adaptive operating ratio $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation, overload, recovery, or regime change; not a universal constant
Λ	Life Number or capstone surplus ratio where explicitly defined	Reserved for Part II-derived life-number notation or synthesis-level surplus measures, never an undefined decoration

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Accumulation or reinforcement coefficient	Sets how fast a pathway, constraint, habit, cascade, or retained structure grows under load
β	Relaxation, decay, clearance, or washout coefficient	Sets loss, recovery, damping, forgetting, degradation, or return toward baseline
γ	Coupling, diffusion, redistribution, or transfer coefficient	Sets spread across nodes, layers, tissues, markets, infrastructures, media, or physical phases
δ, ϵ	Perturbation, tolerance, small offset, or numerical guard	Used only as local thresholds, stability margins, measurement tolerances, or stress increments
η, λ, μ, ν	Efficiency, rate, baseline, intensity, or frequency terms	Meaning is fixed by the equation, subscript, and domain-specific proxy in the paragraph where the term appears
ρ, σ, τ, ϕ	Density/correlation, variability/noise, time-scale, phase, or field terms	Local coefficients only; subscripts bind them to the named pathway, domain, or experiment
Ω, Λ	Domain/state space; Life Number where defined	Ω names a modeled state space; Λ carries life-number or synthesis notation only when the paper defines it

1 Series Context

Part I sets the flow-constraint language used here [3]. Part II develops the Life Number and the origins-of-life use of the same notation [4]. Part III applies AFL to biology and medicine

[5]. Part IV [6], DOI <https://doi.org/10.5281/zenodo.20395901>, is the cross-domain stress test: it asks where the notation clarifies measurement, and where it becomes only a metaphor.

2 Introduction

2.1 The Mujjabi Universal Capacity Law

Building directly upon the Fundamental Physics (Part I) [3], the **Mujjabi Universal Capacity Law** proposes that across all scales—from black hole event horizons to global financial markets and power grids—driving high flux (Φ) against a rigid topological constraint (C) can drive system saturation. When Φ/C approaches critical thresholds, the system does not scale linearly; it undergoes a topological phase transition.

2.2 The Mujjabi Universal Memory Law

The **Mujjabi Universal Memory Law** frames persistent systemic collapse. When a complex network fails to clear its structural memory (M_s) and its relaxation parameter decays ($\tau_{relax} \rightarrow \infty$), temporary stressors become permanent rigidities. This is the proposed CDFD mechanism behind ecological death, economic depressions, and infrastructural gridlock.

The global economy is a singular, highly integrated fluid network. Capital, technology, and raw materials naturally seek to flow from regions of surplus to regions of deficit, driven by the thermodynamic gradient of profit and necessity.

However, nations frequently intervene in this natural flow for geopolitical reasons. Tariffs, embargoes, and sanctions act as artificial dams placed across the river of global commerce. In CDFD, a sovereign state implementing a tariff is actively manipulating the topological constraints of the global active surface.

3 The Geoeconomic Adaptive Surface

We govern bilateral and global trade using the Universal Adaptive Ratio:

$$\Psi_s = \left(\frac{\Phi}{C} \right) \cdot S \cdot M_s \quad (1)$$

For a specific trade corridor between two nations:

- **Flux (Φ):** The monetary value and physical volume of exports and imports crossing the border.
- **Constraint (C):** Tariffs, quotas, regulatory misalignment, and outright economic sanctions.
- **Responsiveness (S):** The elasticity of the domestic economy to substitute imports, find alternative trading partners, or pivot domestic production.

- **Memory (M_s):** Decades-long trade agreements, entrenched supply chains, and specialized infrastructure (e.g., specific pipelines or localized shipping hubs). High memory locks the economy into dependency on a specific partner.

3.1 Trade Wars and Topological Severing

In a free-trade era, tariffs are low, and Φ is extremely high. The global system is highly integrated and operates in a sustained growth regime ($\Psi_s > 1$).

If a trade war begins, nation A applies a massive tariff against nation B. The constraint C spikes artificially. If nation B's economy is highly specialized and dependent on this specific corridor (high memory M_s), it cannot easily pivot. The adaptive ratio for that specific trade sector crashes to $\Psi_s \ll 1$.

The flow Φ of critical components ceases. The sanctioned or tariffed nation enters an ischemic economic state. To survive, the nation must drastically increase its internal responsiveness S —forcing expensive domestic production or forging entirely new, inefficient trade corridors with secondary nations. The artificial constraint forces a massive, thermodynamically inefficient topological rewiring of the global surface.

4 Measurement Layer

The first job in any domain is to choose proxies before drawing conclusions. Φ may be a rate of material, energy, information, capital, or signal flow. C is the bottleneck that slows or redirects that flow. S measures the system's ability to reconfigure under stress, and M_s records the past load that still changes present behavior. A useful application gives those four terms observable definitions, a time scale, and a failure condition. If a standard domain model explains the same data more cleanly, the CDFD/AFL mapping should give way.

4.1 Current Runtime Diagnostic: Universal Network Cascade

The release-local discovery script keeps this paper tied to the current CDFD Runtime [7], including RK4 field updates, surface dynamics, structural-memory diffusion, and a finite-value audit. In the 2026-06-04 runtime pass, the localized hub-drive stress test ended with hub $\Psi_s = 5.21$, peak network $\Psi_s = 5.21$, 21 nodes above $\Psi_s > 1.5$, and 37 maximum memory-locked nodes. I treat those numbers as a runtime sanity check and as a target for later domain measurement, not as a substitute for field data.

5 Major Universal Upgrade: Mechanism, Scale, and Tests

5.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For Trade Networks, the proposed drive is goods, services, capital, information, and contractual demand. The active limitation is ports, tariffs, logistics, finance, and network dependencies. Responsiveness is carried by substitution, inventory, rerouting, diplomacy, and production change,

while retained state is represented by contracts, specialization, infrastructure, and disruption history. The outcome to explain is trade volume, shortages, prices, and recovery.

Table 1: Operational CDFD/AFL map for Paper C-03.

Term	Paper-specific observable meaning	Measurement question
Φ	goods, services, capital, information, and contractual demand	What rate, load, gradient, or demand enters the focal system?
C	ports, tariffs, logistics, finance, and network dependencies	Which measured bottleneck limits transfer, function, or recovery?
S	substitution, inventory, rerouting, diplomacy, and production change	Which process changes effective capacity or reroutes the load?
M_s	contracts, specialization, infrastructure, and disruption history	Which retained state makes equal present loads produce different futures?
Y	trade volume, shortages, prices, and recovery	Which outcome is predicted out of sample, with uncertainty?

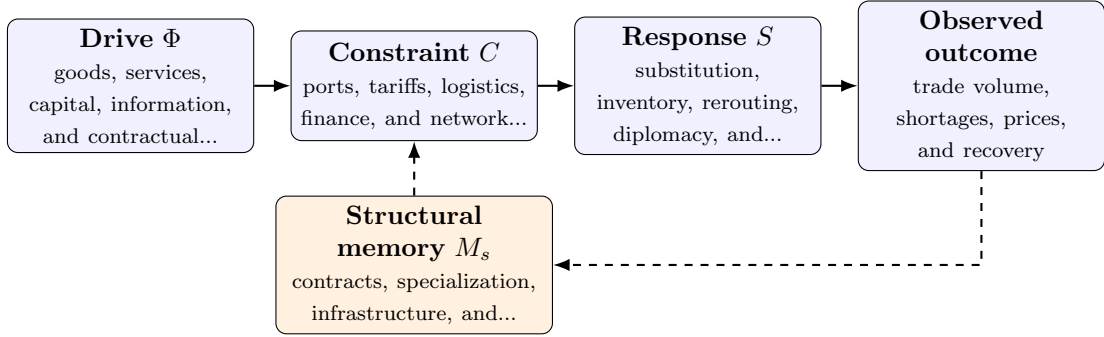


Figure 1: Paper C-03 observable CDFD/AFL architecture for Trade Networks. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

5.2 Minimal Coupled Model

A paper-level model must separate throughput, constraint accumulation, response, and history. One deliberately minimal form is

$$Y(t) = \frac{\Phi(t)S(t)}{\epsilon + C(t)}, \quad (2)$$

$$\frac{dC}{dt} = \alpha\Phi(t) - \beta S(t)C(t) + \gamma\mathcal{L}C(t), \quad (3)$$

$$\frac{dM_s}{dt} = \eta C(t) - \mu M_s(t), \quad (4)$$

$$\Psi_s(t) = \frac{\hat{\Phi}(t)}{\epsilon + \hat{C}(t)} \hat{S}(t) [1 + \omega \hat{M}_s(t)]. \quad (5)$$

Here Y is the measured output, $\epsilon > 0$ prevents a singular denominator, $\mathcal{L}$ is a spatial or network coupling operator, and hats denote explicitly normalized variables. The sign and magnitude of ω must be estimated: memory can preserve useful organization or amplify damage. No fixed threshold is universal before the variables, normalization, and observation scale are declared.

For this paper, the hypothesized causal sequence is specific. A change in goods, services, capital, information, and contractual demand arrives faster than substitution, inventory, rerouting,

diplomacy, and production change can compensate. This raises or redistributes ports, tariffs, logistics, finance, and network dependencies. If the load persists, the retained state encoded by contracts, specialization, infrastructure, and disruption history changes the next response, so a later event of equal magnitude can produce a different trajectory in trade volume, shortages, prices, and recovery. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

5.3 Cross-Scale Universality Without Mechanistic Erasure

For the socioeconomic systems family, the universal comparison is between human and institutional throughput, unequal access to capacity, adaptive coordination, and historical path dependence. The variables are not morally neutral: aggregation can hide who receives flow, who bears constraint, and whose prior losses become persistent memory. [1, 9, 2, 8] The CDFD programme is cumulative: Part I supplies the flow–constraint foundation [3]; Part II asks when maintained throughput supports persistent organized states [4]; Part III develops adaptive limitation and biological memory [5]; and Part IV [6] tests how much of that architecture survives translation. The CDFD Runtime [7] supplies a reproducible numerical stress surface, but it does not substitute for the domain evidence named here.

The required scale declaration for this paper is days to decades; firms to the world economy. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

5.4 Evidence Ladder and Discriminating Tests

Table 2: Evidence ladder for Paper C-03. Each rung can fail independently.

Test	Design	Discriminating result
Proxy audit	Measure customs data, input-output tables, shipping traces, prices, and disruption records and preregister how each record maps to Φ , C , S , M_s , and Y .	The mapping fails if proxies are non-identifiable, circular, or change meaning across cases.
Load ramp	Apply or observe a corridor closure or supplier loss during high demand while estimating the dominant constraint and response time.	A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint.
Recovery and memory	Match present load across units with different histories, then compare recovery paths.	The memory claim requires history-dependent divergence after current conditions are controlled.
Model comparison	Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data.	The paper’s stated falsifier is: network memory does not improve shortage or trade-recovery prediction.

The strongest practical design combines a prospective perturbation with a recovery window. The load-ramp phase estimates where constraint begins to rise faster than output. The recovery phase estimates β and μ , separating fast relaxation from persistent memory. A topology

or coupling intervention then asks whether rerouting changes the cascade as predicted. Null results are informative because they identify domains in which the universal notation adds no measurable value.

5.5 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from customs data, input-output tables, shipping traces, prices, and disruption records and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL state extension and test whether it improves prediction of trade volume, shortages, prices, and recovery. Fourth, repeat the analysis across the declared scale range and publish negative cases as part of the universality audit.

Three boundaries are non-negotiable. Correlation between flow and failure does not identify a constraint mechanism. A fitted memory term does not prove that the stored state has the same material meaning in another domain. Finally, a runtime cascade is evidence about the implementation, not evidence that the real system follows it. The paper becomes stronger when it can say precisely where the translation stops.

6 Conclusion

Trade wars and sanctions are not merely political tools; they are the intentional weaponization of thermodynamic constraints. By framing international trade within CDFD, we see that global commerce behaves as an adaptive biological membrane. Artificial geopolitical constraints force traumatic structural adaptations, supporting the testable hypothesis that economic isolation can drive either violent internal restructuring or systemic decay.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Part-specific runtime diagnostics are under each Part's `outputs/` directory.

Release-wide code: `Part_E_Synthesis/supplementary/`.

Generated summaries: `Part_E_Synthesis/outputs/`.

Figures: `Part_E_Synthesis/figures/`.

7 Reference Layer

References

- [1] Albert-Laszlo Barabasi and Reka Albert. Emergence of scaling in random networks. *Science*, 286(5439):509–512, 1999. DOI: <https://doi.org/10.1126/science.286.5439.509>.
- [2] C. S. Holling. Resilience and stability of ecological systems. *Annual Review of Ecology and Systematics*, 4:1–23, 1973. DOI: <https://doi.org/10.1146/annurev.es.04.110173.000245>.
- [3] Steve Bico Mujjabi. Cdfd part i: Fundamental physics - constraint-driven field theory and flux dynamics, 2026. DOI: <https://doi.org/10.5281/zenodo.20250821>.
- [4] Steve Bico Mujjabi. Cdfd part ii: Origins of life and tri-regime bioenergetics - constraint-driven flux dynamics, 2026. DOI: <https://doi.org/10.5281/zenodo.20264779>.
- [5] Steve Bico Mujjabi. Cdfd part iii: Adaptive flux limitation biology and medicine - constraint-driven flux dynamics, 2026. DOI: <https://doi.org/10.5281/zenodo.20344695>.
- [6] Steve Bico Mujjabi. Cdfd part iv: Universal adaptive flux limitation - constraint-driven flux dynamics, 2026. DOI: <https://doi.org/10.5281/zenodo.20395901>.
- [7] Steve Bico Mujjabi. Cdfd runtime: Constraint-driven flux dynamics and cdf execution engine, 2026. DOI: <https://doi.org/10.5281/zenodo.20343160>.
- [8] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948. DOI: <https://doi.org/10.1002/j.1538-7305.1948.tb01338.x>.
- [9] Duncan J. Watts and Steven H. Strogatz. Collective dynamics of small-world networks. *Nature*, 393:440–442, 1998. DOI: <https://doi.org/10.1038/30918>.